

Smart City Healthcare – Diabetic Retinopathy (DR) Screening with Deep Learning

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Abstract & Objective

Diabetic Retinopathy (DR) is a severe complication of diabetes that damages the retina. It is a leading cause of preventable blindness, requiring timely and accurate screening

However, it is an ongoing challenge to correctly classify fundus images into correct DR classes (**0**: Normal, **1**: Mild non-proliferative DR (NPDR), **2**: Moderate NPDR, **3**: Severe NPDR, **4**: Proliferate NPDR) as images have to handle extreme lighting variations, blurring, camera differences, and severe class imbalance.

With deep learning, particularly **Convolutional Neural Networks (CNNs)** and **Vision Transformers (ViTs)**, machine learning models can now learn the subtle differences that separates the different classes of DR, allowing screening to be automated, reducing the diagnostic burden on ophthalmologists.

Our approach on this problem is an optimized preprocessing pipeline that can uses **Contrast Limited Adaptive Histogram Equalization (CLAHE)**, **Graham's Method** and **Feature Adaptive Cropping**. Then, images are processed through an stacking ensemble that consists of **ResNet-34** and **EfficientNet-B4**, using a **Random Forest** meta-classifier to produce the class. The model is trained on the full provided dataset, using **Macro-F1** to evaluate the model.

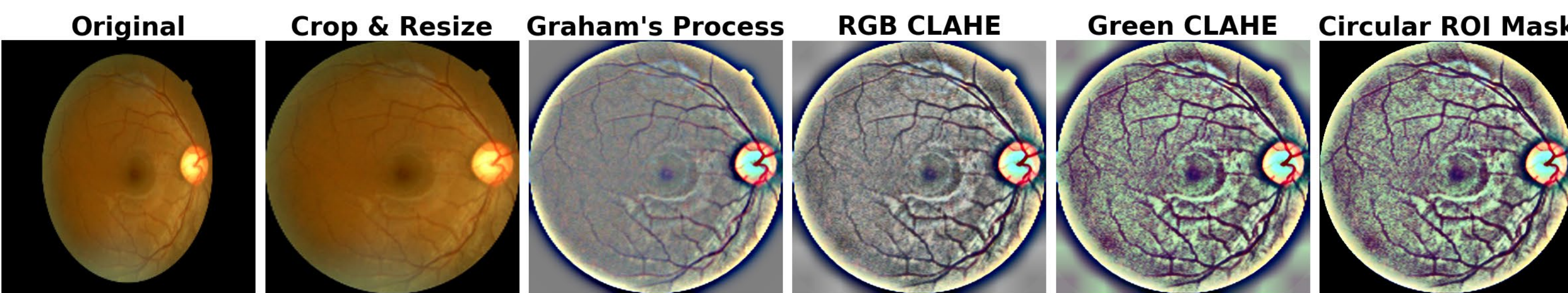
Methodology

Preprocessing

Preprocessing pipeline is applied to all images before passing through the models as a **384x384 RGB image**.

This includes:

1. **Adaptive cropping and resize**: Remove black borders
2. **Graham's Process**: Adding Gaussian noise to remove lighting variations etc.
3. **Contrast Limited Adaptive Histogram Equalization (CLAHE)**: Enhance contrast on different channels of the image
4. **Circular ROI Mask**: Remove aberrations



Training Methods

A few models are trained on the provided data, including vanilla **CNNs**, **ViTs**, **EfficientNet-B4** and **CatBoost**.

Training is done on 2 **T4 GPUs** with **AdamW** (**learning rate**: 2e-4, **weight decay**: 1e-5) with **gradient norm clipping**, **gradient scaling** and **learning rate scheduler** to stabilize training. Training utilizes **Cross Entropy Loss** as we found it to stabilize training better than **Focal Loss**.

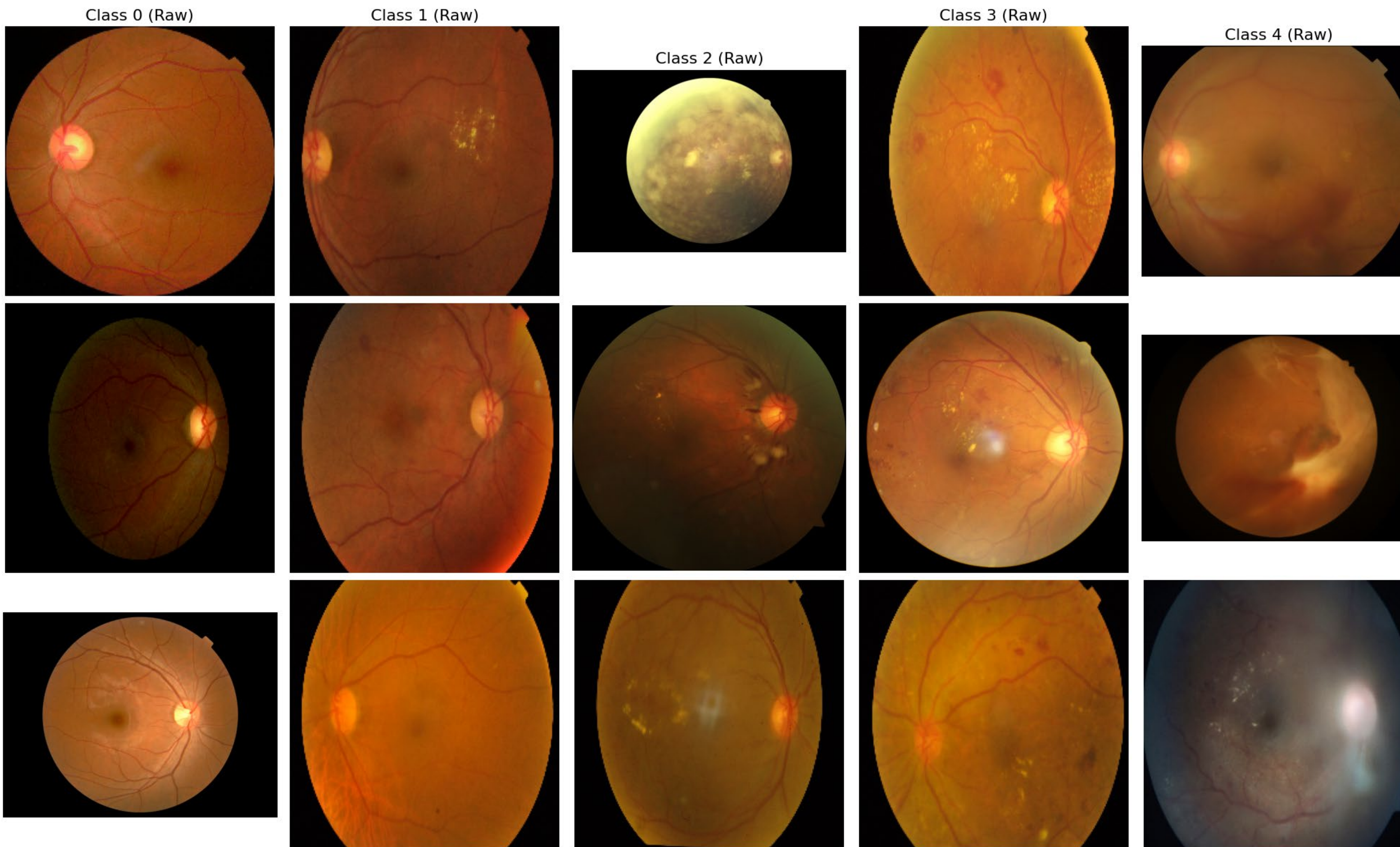
For each epoch, an external validation set is monitored and model is saved if **validation F1** score improves.

After training, all model combinations are evaluated on an external test set.

Rank	Model Combination	Stacking Method	Test Accuracy	Test Macro-F1
1	ResNet-34 + EfficientNet-B4 + CatBoost	Logistic Regression	0.8858	0.8791
2	ResNet-34 + EfficientNet-B4 + ViT	Random Forest	0.8845	0.8778
3	ResNet-34 + EfficientNet-B4	Random Forest	0.8885	0.8767
4	ResNet-34 + EfficientNet-B4	Logistic Regression	0.8845	0.8749
5	ResNet-34 + EfficientNet-B4 + ViT + CatBoost	Logistic Regression	0.8832	0.8746

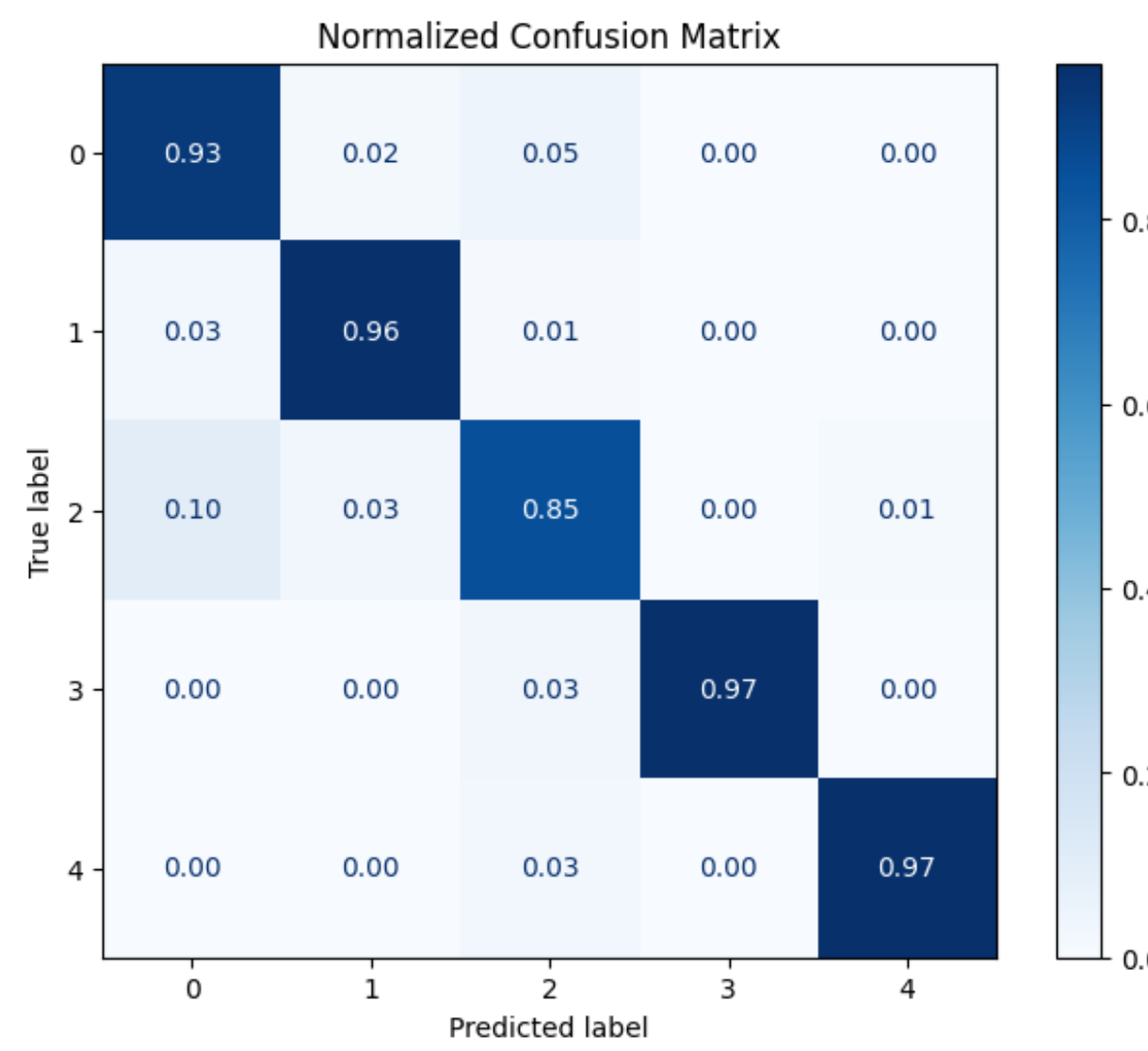
However, based on the inference speed and model size, a **ResNet-34** and **EfficientNet-B4** stacked with **Random Forest** meta classifier is chosen

Raw Data



Results

On the external test set, the final model is evaluated. Below is the normalized confusion matrix and classification report.



Class	Precision	Recall	F1-Score	Support
0	0.93	0.93	0.93	6266
1	0.71	0.96	0.82	630
2	0.92	0.85	0.88	4477
3	0.94	0.97	0.95	236
4	0.93	0.97	0.95	913
accuracy			0.91	12522
macro avg	0.88	0.94	0.91	12522
weighted avg	0.91	0.91	0.91	12522

Total parameter count: ~38.8M

Total model size: 153.6 MB

Inference speed: ~25 images per second on NVIDIA RTX 4000 Laptop GPU

Conclusion

This project successfully demonstrates a highly robust, data-centric approach to Diabetic Retinopathy screening in real-world conditions. By prioritizing advanced image enhancement techniques over exhaustive hyperparameter tuning, the pipeline effectively mitigates extreme lighting variations and camera artifacts. The stacking ensemble, leveraging the fast feature extraction of ResNet-34 and the high capacity of EfficientNet-B4, smoothed by a Random Forest meta-classifier can captures complex, non-linear disease patterns without overfitting. Crucially, by optimizing for the Macro F1-score rather than raw accuracy, the model explicitly penalizes majority-class bias and ensures the clinical safety of detecting rare but severe proliferative DR cases.



DEMO QR



CODE QR

References

1. He, K., Zhang, X., Ren, S., & Sun, J. (2016). "Deep Residual Learning for Image Recognition." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*
2. Tan, M., & Le, Q. (2019). "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks." *Proceedings of the 36th International Conference on Machine Learning (ICML)*
3. Graham, B. (2015). "Kaggle Diabetic Retinopathy Detection Competition Report."
4. Zuiderveld, K. (1994). "Contrast Limited Adaptive Histogram Equalization." *Graphics Gems IV*, San Diego: Academic Press Professional

Preprocessing

CNN Model
Logits

Ensemble
Prediction